

Step 5 One-Pager — Neural Networks for Imperfect-Information Games

Research on the possibilities for applying Artificial Intelligence in computer games

Alexander Andreev

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Problem. Steps 2-4 all store a strategy as a table indexed by information set, and Step 4 showed both escapes from that table — a better solver, a smaller game — running out. The third escape is to stop enumerating: replace the regret table with a function approximator that *generalises* across similar information states, at fixed memory. Step 5 studies that substitution — Deep CFR, its single-network variants, NFSP — and what it costs in guarantees.

Approach. Scope, stated honestly: **this step stopped after exploration and reading — there is no from-scratch Phase-4 implementation and no implementation report**, and the survey’s closing synthesis is still a stub. The measured work is a two-day exploration against OpenSpiel’s reference solvers on Leduc and Kuhn — Deep CFR at three network sizes, NFSP on both games, tabular MCCFR and CFR+ as baselines — graded by OpenSpiel’s own exploitability so the head-to-head is honest (Step 3’s MCCFR runs on a custom engine and is not comparable). **All numbers below are measured** (implementation/step05/exploration/logs/day0{1,2}_results.json).

Key results (measured).

- *The step’s most actionable artifact is a library bug.* OpenSpiel 1.6.12’s PyTorch DeepCFRSolver **never trains its advantage networks**: if `len(samples.info_state == 0)` takes `len` of an elementwise comparison, always truthy, so it returns before the optimiser step. Advantage losses come back silently `None` and exploitability sits at random-strategy level (**~1.69**) whatever the iteration count. A one-character fix, patched locally.
- *Where the table fits, the table wins — decisively.* Leduc at comparable wall time: CFR+ **0.00123** (400 iter, 92 s), tabular MCCFR **0.0967** (50k iter, 68 s), Deep CFR **1.49-1.70** (120 iter, ~95 s), NFSP **2.46** (50k episodes) — CFR+ roughly **1,400x** better than Deep CFR.
- *Those neural numbers are budget artifacts, not verdicts.* Deep CFR’s curves are flat from iteration 30 to 120 where the paper uses 400+, and OpenSpiel’s own NFSP Leduc example runs **2e7** episodes — **400x** our budget. Rule adopted: distrust an NFSP Leduc curve below ~1e6 episodes.
- *Smaller network won at this budget.* (32,32) **1.1441** beat (64,64) **1.7002** and (128,128,128) **1.4908** — consistent with underfitting on few distinct samples, and expected to reverse with more iterations.
- *What under-training looks like inside the policy.* Against a 400-iteration CFR+ reference, Deep CFR at 40 iterations has **median total-variation 0.35** across 8 sampled Leduc info states (range 0.163-0.708): the direction of each decision is usually right, but the probabilities are pulled toward uniform (**0.73/0.27** where CFR+ is **0.92/0.07**) — MSE regression on noisy counterfactual samples buys an under-confident smoothing of equilibrium.
- *The cost structure that justifies the family anyway.* A Deep CFR outer iteration costs **~0.7 s** against a tabular MCCFR iteration’s **~1.4 ms** (~500x), but its per-iteration work is roughly constant in game size while the tabular cost scales with information-set count. Leduc is a teaching benchmark, where neural methods lose to baselines they were never meant to beat.

Thesis connection. Deep CFR’s advantage network is Step 1’s value network with a counterfactual target — the joint that lets Steps 6-8 leave table-sized games behind. More pointedly, the under-confident smoothing measured here is the failure mode Step 7 later prices: a model with the right direction and the wrong frequency is what makes best-responding to an imperfect read lose to Nash.

Open questions. The step’s own gate is unmet: a hand-coded Deep CFR with a Vitter reservoir sampler, and DREAM’s outcome sampling with baseline subtraction, both remain outstanding, so “can I build it” is unanswered. Untested: whether the tabular-versus-neural ordering flips exactly at Step 3’s node-count crossover, and whether the info-state tensor’s structure really matters more than network depth.